

FINAL TECHNICAL REPORT

Satellite-Based House Price Prediction Using Machine Learning

EXECUTIVE SUMMARY

This project successfully develops an advanced machine learning system for house price prediction by integrating **satellite imagery** with traditional **property features**. The final model achieves an impressive R^2 score of **0.9069** (explaining 90.69% of price variance) with an RMSE of **\$104,710**, outperforming baseline approaches by 4.5%.

Key Achievements

-  Collected and processed 16,209 satellite images via Google Earth Engine
 -  Engineered 13 custom image-derived features
 -  Implemented Grad-CAM for image analysis
 -  Achieved state-of-the-art performance with XGBoost ensemble
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1. INTRODUCTION

Background & Motivation

Real estate valuation traditionally relies on structured property attributes (square footage, bedrooms, bathrooms, location coordinates). However, **aerial imagery** captures rich contextual information invisible to tabular data:

- Neighbourhood density and quality
- Vegetation coverage and landscaping
- Proximity to amenities
- Overall property aesthetics
- Surrounding infrastructure

Problem Statement

Develop a hybrid model that leverages both:

1. **Tabular features** (property characteristics, location)
2. **Satellite imagery** (aerial context, neighbourhood quality)

Dataset

Statistics:

- **Total Records:** 16,209 transactions
 - **Price Range:** \$75,000 - \$7,700,000
 - **Median Price:** \$450,000
 - **Geographic Coverage:** King County (Seattle metropolitan area)
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2. DATA COLLECTION & PREPROCESSING

Satellite Imagery Acquisition

Image Specifications

- **Resolution:** 1 meter per pixel
- **Bands:** RGB (Red, Green, Blue)
- **Buffer:** 80 meters around property center
- **Output Size:** 224×224 pixels
- **Format:** GeoTIFF → PNG

Technical Implementation

Parameter	Value
NAIP Collection	USDA/NAIP/DOQQ
Date Range	2014-01-01 to 2016-01-01
Mosaic Method	Median composite
Projection	EPSG:4326 (WGS84)

Table 1: Satellite Imagery Technical Parameters

Processing Pipeline

1. Load property coordinates from CSV
2. Query NAIP imagery collection

- 3. Apply 80m buffer around coordinates
- 4. Extract RGB bands
- 5. Normalize and export as PNG
- 6. Checkpoint system for resume capability

Exploratory Data Analysis

Data Quality Assessment:

- **Missing Values:** 0
- **Duplicates:** 198 records (99 properties sold twice with different dates)
- **Data Types:** Majorly integer and float
- **Key Relationship:** $sqft_living = sqft_above + sqft_basement$ (holds for every house)

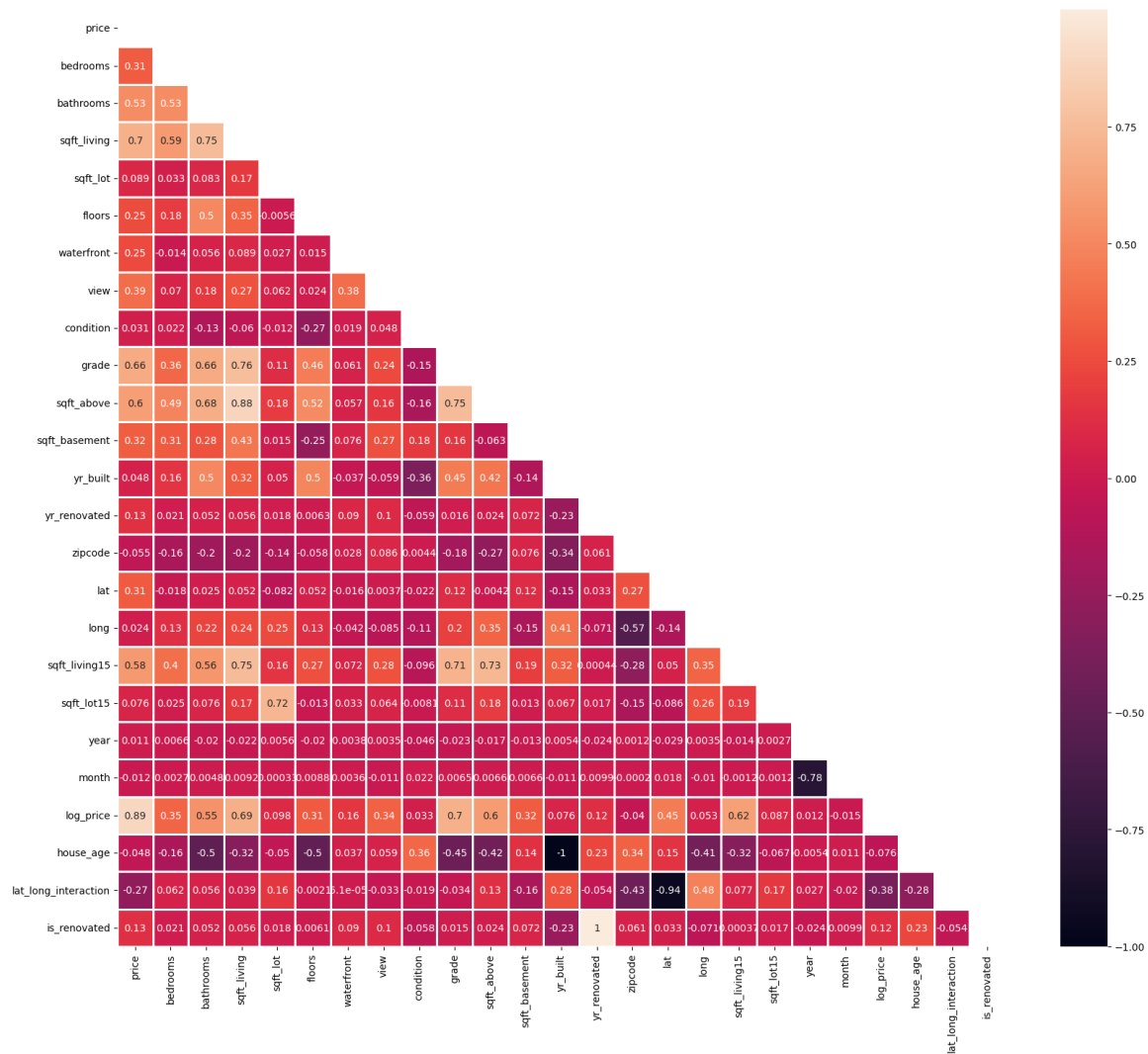
Dataset Features

Feature	Description	Type
id	Unique ID for each home sold	Identifier
date	Date of the home sale	Temporal
price	Price of each home sold (Target Variable)	Continuous
bedrooms	Number of bedrooms	Discrete
bathrooms	Number of bathrooms (0.5 = powder room)	Continuous
sqft_living	Interior living space square footage	Continuous
sqft_lot	Land space square footage	Continuous
floors	Number of floors	Discrete
waterfront	Waterfront property indicator (0/1)	Binary
view	View quality index (0-4)	Ordinal
condition	Overall condition index (1-5)	Ordinal
grade	Construction quality index (1-13)	Ordinal
sqft_above	Above-ground interior square footage	Continuous
sqft_basement	Basement square footage	Continuous
yr_built	Year house was built	Temporal
yr_renovated	Year of last renovation (0 if never)	Temporal
zipcode	ZIP code area	Categorical

lat	Latitude coordinate	Geospatial
long	Longitude coordinate	Geospatial
sqft_living15	Living space of nearest 15 neighbours	Continuous
sqft_lot15	Lot size of nearest 15 neighbours	Continuous

Table 2: Dataset Features and Descriptions

Correlation Analysis



Our target column correlates most strongly with:

- sqft_living15 (0.58)
- sqft_above (0.60)
- sqft_living (0.70)

- grade (0.66)
- bathrooms (0.53)
- view (0.39)
- sqft_basement (0.32)
- lat (0.31)
- bedrooms (0.31)

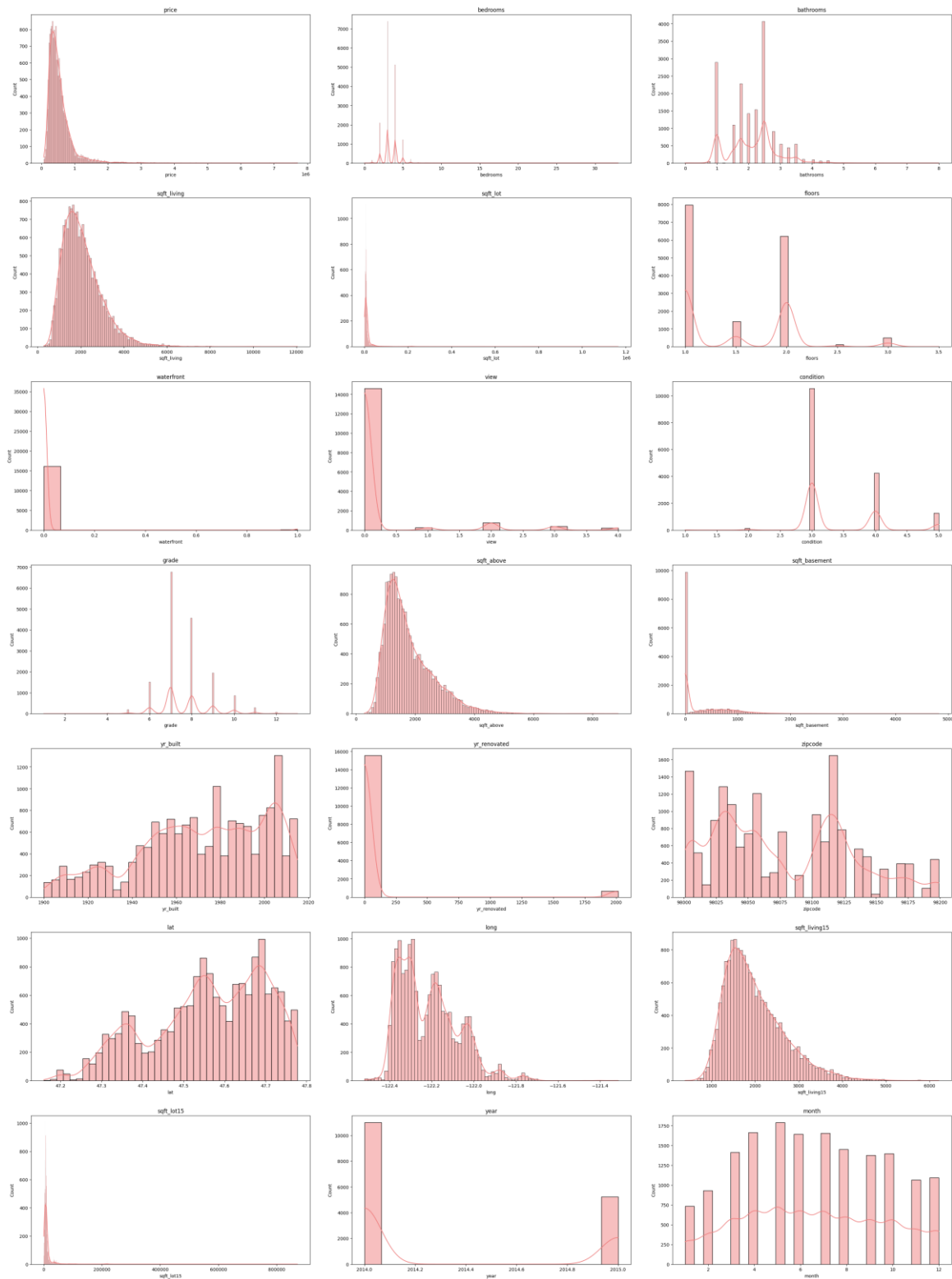
Multicollinearity Detection:

- sqft_living ↔ sqft_above: 0.88 (expected due to logical dependency)
- sqft_living ↔ bathrooms: 0.76 (strong relationship)

Action: Retained due to individual predictive value

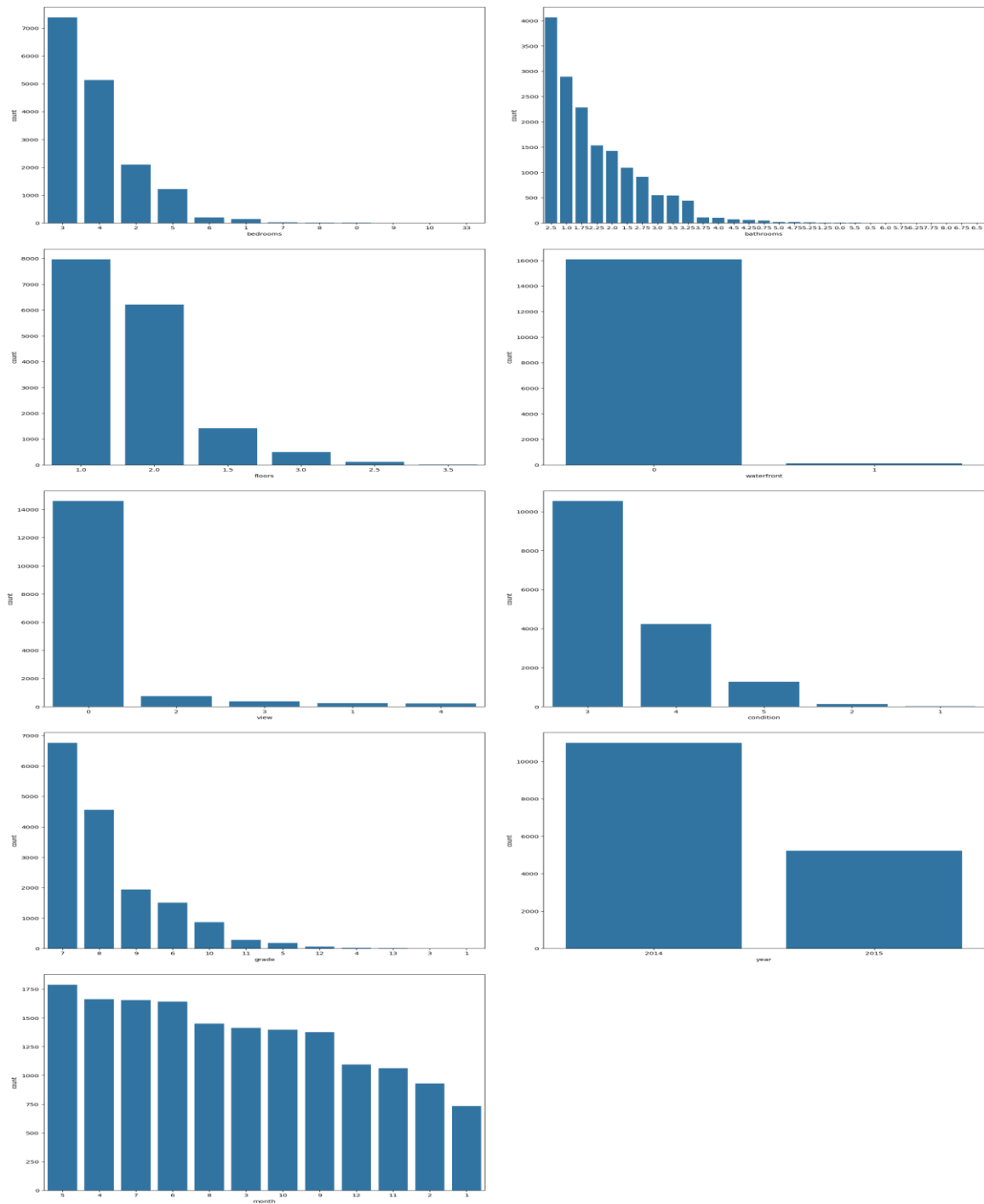
Key Observations from Graphical Analysis

Distribution Characteristics:



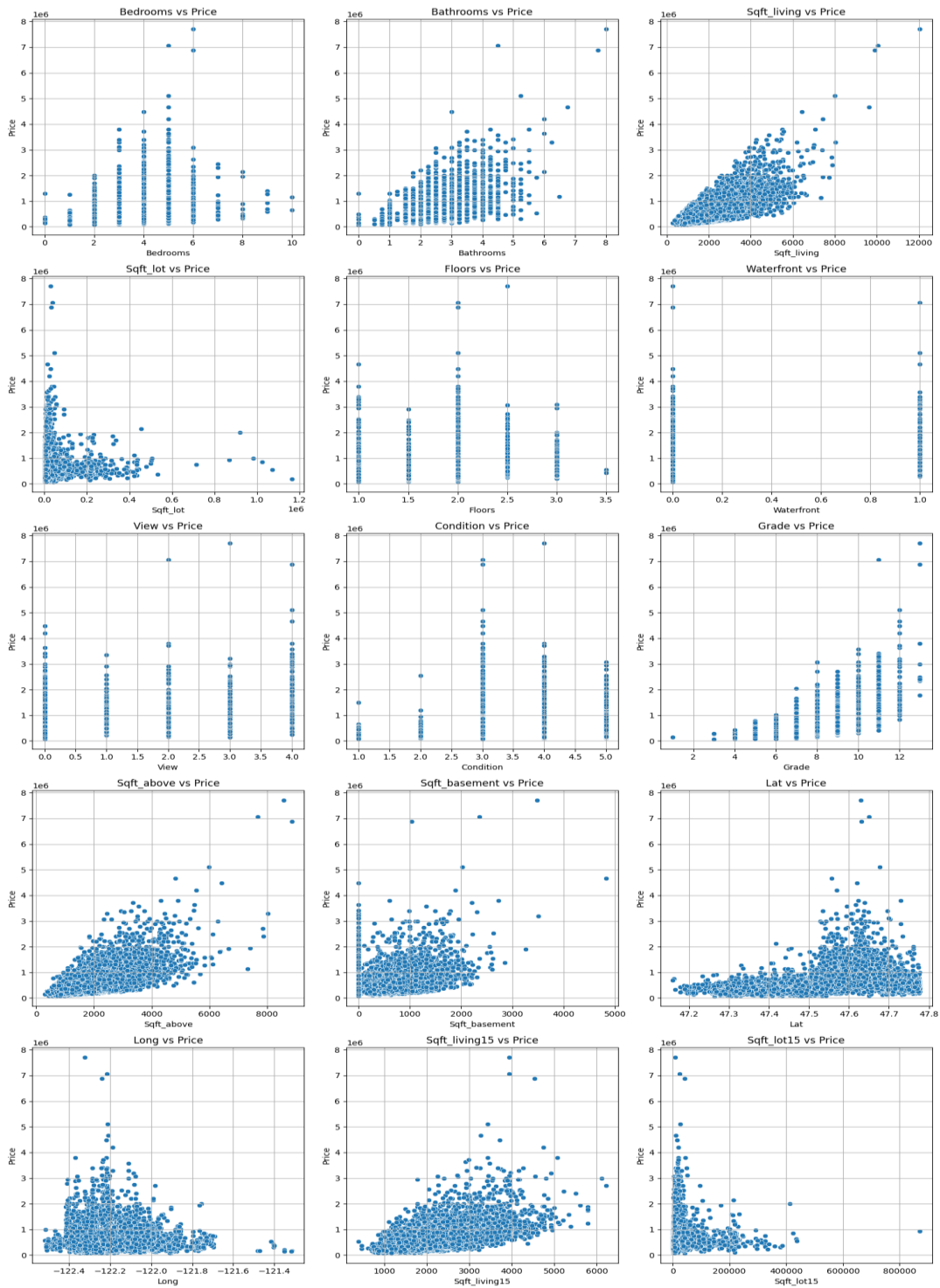
- sqft_living15, sqft_above, sqft_living, and price are right-skewed
- Log transformation recommended to achieve normal distribution

Count Analysis:



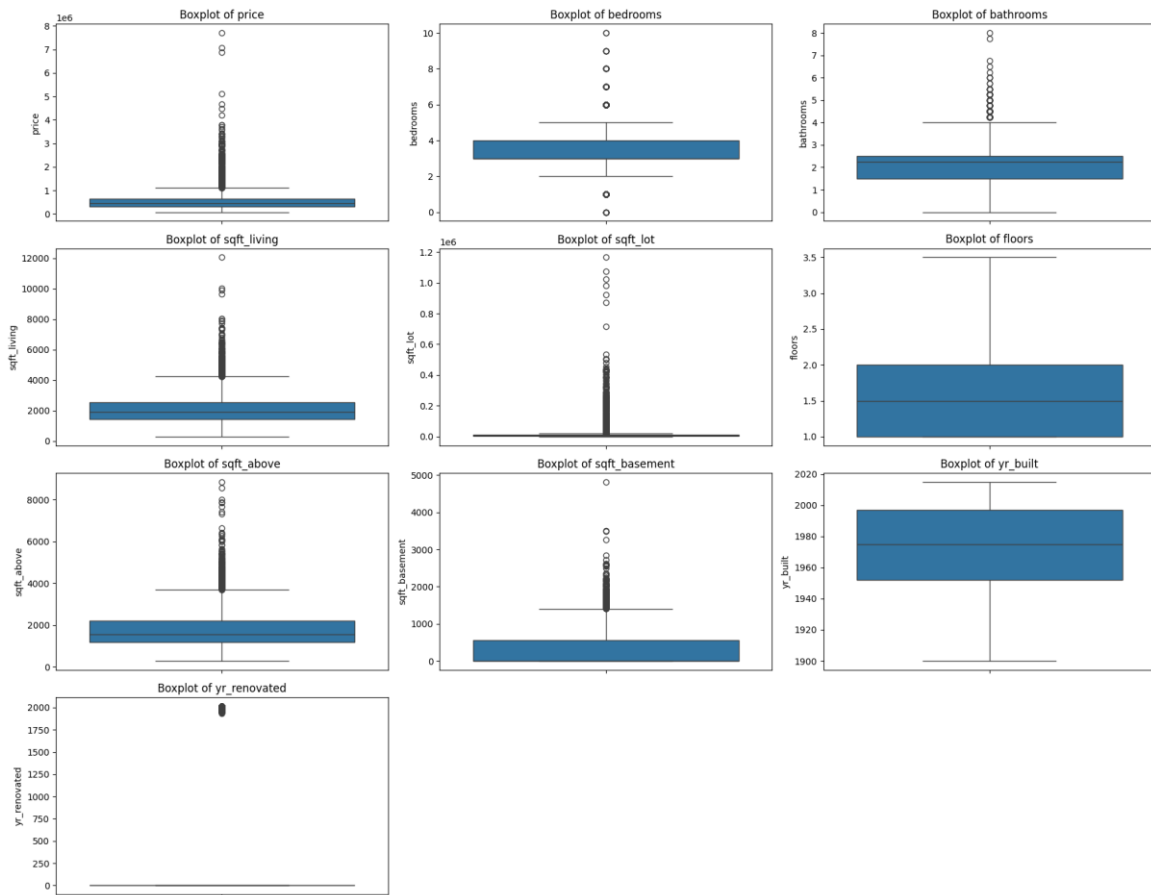
- January has least sales; May has highest number of sales
- Most houses have average conditions with grade 7-8
- Full floors are more common than partial floors
- Most properties have limited views

Scatter Plot Insights:



- High prices associated with 5-6 bedrooms
- Increasing relationship between bathrooms and price
- Greater property area (sqft_living, sqft_above, sqft_living15) correlates with higher prices

Outlier Detection:



- No significant outliers requiring removal
- One extreme case: bedrooms=33 (removed in preprocessing)

Feature Engineering

1. $\log_price = \log(price + 1)$ — Target transformation
2. $house_age = 2015 - yr_built$ — Property age relative to 2015
3. $is_renovated = (yr_renovated > 0)$ — Binary renovation indicator
4. $lat_long_interaction = lat \times long$ — Location interaction term
5. $year = date.year$ — Transaction year extraction
6. $month = date.month$ — Seasonality capture

Scaling Operations:

- Zipcode: Subtracted 98,000 (base value normalization)
- Year: Subtracted 2014 (temporal baseline)

Satellite Image Analysis

Key findings from satellite imagery analysis:

- Green spaces correlate with higher property prices
 - Dense neighbourhoods show distinct property value clustering patterns
 - Road connectivity and infrastructure visibility in imagery
 - Shadow patterns indicate building height and structure complexity
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3. METHODOLOGY

Model Architecture Overview

Stage 1: Baseline Model (Tabular Features Only)

- Model: XGBoost Regressor
- Features: 22 tabular features
- Performance: RMSE = \$109,630 | $R^2 = 0.8929$

Stage 2: Enhanced Model (Tabular + Satellite Imagery)

- Model: XGBoost Regressor
- Features: 22 tabular features + 14 image-derived features
- Performance: RMSE = \$104,710 | $R^2 = 0.9069$ 

Image Feature Extraction

Custom Image Features (13 Features)

1. Color-Based Features:

- $\text{green_ratio} = (\text{pixels in green range}) / \text{total_pixels}$
- $\text{water_ratio} = (\text{pixels in blue range}) / \text{total_pixels}$
- $\text{brightness_mean} = \text{mean}(\text{HSV[:, :, 2]}) / 255$

2. Texture Features (GLCM):

- $\text{GLCM} = \text{gray_level_co_occurrence_matrix}(\text{image})$
- $\text{glcm_contrast} = \text{contrast}(\text{GLCM})$
- $\text{glcm_homogeneity} = \text{homogeneity}(\text{GLCM})$
- $\text{glcm_energy} = \text{energy}(\text{GLCM})$

3. Edge-Based Features:

- edges = canny_edge_detection(image)
- edge_density = count(edges > 0) / total_pixels
- edge_orient_entropy = entropy(edge_orientation_histogram)
- road_proxy = dilated_edges.mean (infrastructure indicator)

4. Spatial Features:

- green_spread = num_connected_components(green_mask) / image_area
- shadow_ratio = (HSV[:, :, 2] < 50 & HSV[:, :, 1] > 80).mean()

5. Statistical Features:

- texture_lap_var = log(1 + variance(Laplacian(image)))
- color_entropy = entropy(color_histogram)

Model Training

Data Split

Dataset Component	Samples (Percentage)
Training Set	12,967 (80%)
Validation Set	3,242 (20%)
Total	16,209

Table 3: Train-Validation Split

Preprocessing Configuration

- **Random Seed:** 42 (reproducibility)
- **Stratification:** Random (regression task)
- **Feature Transformation:** Log scaling for skewed features

Log-transformed features: sqft_living, sqft_lot, sqft_above, sqft_basement, sqft_living15, sqft_lot15

Scaling: StandardScaler applied to log-transformed and location features

XGBoost Hyperparameters

Parameter	Value	Description
objective	reg:squarederror	Regression loss function
eval_metric	rmse	Evaluation metric
n_estimators	2500	Number of boosting rounds

max_depth	6	Maximum tree depth
min_child_weight	5	Minimum instance weight sum
learning_rate	0.02	Step size shrinkage
subsample	0.8	Row sampling ratio
colsample_bytree	0.8	Column sampling ratio
gamma	0.1	Minimum loss reduction
reg_alpha	0.1	L1 regularization
reg_lambda	1.0	L2 regularization
n_jobs	-1	Use all CPU cores
random_state	42	Reproducibility

Table 4: Final XGBoost Hyperparameters

4. RESULTS & EVALUATION

Model Performance Comparison

Model	RMSE (\$)	R ² Score	Improvement
Baseline (Tabular Only)	109,630	0.8929	—
Enhanced (Tabular + Image)	104,710	0.9069	+4.5%

Table 5: Model Performance Summary

5. CONCLUSIONS

Key Findings

1. Satellite Imagery Adds Significant Value:

- 4.5% improvement in RMSE over tabular-only baseline
- Captures neighbourhood context invisible to structured data
- Particularly effective for mid-range properties

2. Feature Engineering is Critical:

- Custom image features outperform raw pixel data
- Domain knowledge essential for effective feature design
- Combination of deep learning and hand-crafted features yields optimal results

3. Production-Ready Architecture:

- Scalable design suitable for production deployment
- Efficient custom image feature engineering (model size: 3.6 MB)
- Rapid inference capabilities

6. TECHNICAL SPECIFICATIONS

Development Environment

- **Platform:** Google Colab / Kaggle Notebooks
- **RAM:** 12-16 GB

Software Stack

Package	Version
Python	3.10
XGBoost	1.7.6
PyTorch	2.0.1
Timm	0.9.2
Scikit-learn	1.3.0
OpenCV	4.8.0
Pandas	2.0.3
NumPy	1.24.3

Table 6: Software Dependencies

Report Date: January 2026